SAE./No.234/April 2023

Studies in Applied Economics

THE EUROZONE: A GOLDEN GROWTH RATE AND CREDIT COUNTERPARTS ANALYSIS

Steve H. Hanke, John Greenwood, & Jingxing Zou

Johns Hopkins Institute for Applied Economics,
Global Health, and the Study of Business Enterprise

The Eurozone: A Golden Growth Rate and Credit Counterparts Analysis

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The *Studies in Applied Economics* series is under the general direction of Prof. Steve H. Hanke, Founder and Co-Director of the Johns Hopkins Institute for Applied Economics, Global Health and the Study of Business Enterprise (hanke@jhu.edu). The views expressed in each working paper are those of the authors and not necessarily those of the institutions' that the authors are affiliated with.

About the Authors

Steve H. Hanke is a Professor of Applied Economics and Founder & Co-Director of The Johns Hopkins Institute for Applied Economics, Global Health, and the Study of Business Enterprise in Baltimore. Prof. Hanke is a well-known currency reformer, and a currency and commodity trader. Prof. Hanke served on President Reagan's Council of Economic Advisers, has been an adviser to five foreign heads of state and five foreign cabinet ministers, and has held a cabinet-level rank as a state counselor in both Lithuania and Montenegro. He has been awarded seven honorary doctorate degrees and is an Honorary Professor at four foreign institutions. He was President of Toronto Trust Argentina in Buenos Aires in 1995, when it was the world's best-performing mutual fund. Currently, he serves as Chairman of the Supervisory Board of Advanced Metallurgical Group N.V. in Amsterdam. In 1998, he was named one of the twenty-five most influential people in the world by *World Trade Magazine*. In 2020, Prof. Hanke was named a Knight of the Order of the Flag.

John Greenwood was until December 2021 the Chief Economist of Invesco Ltd., a global asset management company. As editor of the Asian Monetary Monitor over the period 1977-96, he is widely credited as the designer of the restored currency board system in Hong Kong at the time of the currency crisis in 1983, a model for numerous subsequent currency board systems. Holding an MA and Honorary Doctorate from the University of Edinburgh, he is the author of "Hong Kong's

Link to the US Dollar – Origins and Evolution” (Hong Kong University Press, 2007 and 2022). He has been a Fellow at the Johns Hopkins Institute for Applied Economics, Global Health, and the Study of Business Enterprise since June 2016.

Jingxing Zou (jzou12@jhu.edu) is a sophomore at the Johns Hopkins University in Baltimore, pursuing majors in applied mathematics and statistics and international studies, with a minor in economics. He contributed to this paper as the Chief of the Monetary Analysis Team at the Institute for Applied Economics, Global Health, and the Study of Business Enterprise in Spring 2023. He will graduate in May 2025.

Abstract

This paper presents a Golden Growth Rate (GGR) analysis of the eurozone for the period 2010 through 2019. A credit counterpart analysis for the period 2010 through 2022 is also presented with the purpose of understanding the drivers of the eurozone’s monetary growth.

SECTION 1: Golden Growth Analysis[1]

The GGR is the growth rate of the money supply that is consistent with hitting a defined inflation target. The following determination of the eurozone's GGR covers the period of 2010 through 2019; to avoid disruptions caused by the COVID-19 pandemic, the years thereafter are omitted.

To calculate the GGR, we utilized the income form of the Quantity Theory of Money (QTM), $MV = Py$, where M denotes the money supply; V denotes the average number of times *per unit time* that the money stock is used in making income transactions; P denotes the price level; and y denotes real GDP (Py denotes nominal GDP). To operationalize the QTM, we used its percent-change form: $\Delta M + \Delta V = \Delta P + \Delta y$.[2]

To solve for the golden growth rate, we rearranged the equation to $\Delta M = \Delta P + \Delta y - \Delta V$, where ΔM represents the golden growth rate of the eurozone's broad money supply, M3.[3] In this scenario, because we are solving for the golden growth rate that is consistent with the eurozone's inflation target, ΔP is the inflation target set by the ECB.[4] Δy is the eurozone's average YoY change in real GDP[5] over the period. To obtain ΔV, for each year in the relevant time span (in this case, 2010-2019), we first divide the nominal GDP[6] by the money supply to obtain the money velocity, $V = \frac{End\ of\ year\ Nominal\ GDP}{End\ of\ year\ M3}$. Then, we created a time series of yearly money velocity percent changes with $\Delta V(t) = \frac{V(t)-V(t-1)}{V(t-1)}$. Taking the average of all $\Delta V(t)$ for all t in the relevant time span reduces the noise in the result. Thus, the final number, ΔV, is an average of money velocity changes throughout the years.

Between 2010 and 2019, the eurozone's annual inflation target set by the ECB was constant at 2%. In the same period, the eurozone's annual real GDP growth averaged 1.4%, and the average annual rate of change in velocity was -1.2%. Thus, the percentage-change-form of the Quantity Theory, $\Delta M = \Delta P + \Delta y - \Delta V$, determines that the eurozone's golden growth rate for M3 is *2%* +

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[1] For treatment of the Quantity Theory of Money as it relates to Golden Growth Rates, see Greenwood-Hanke (2021) https://onlinelibrary.wiley.com/doi/abs/10.1111/jacf.12479

[2] For a derivation of the percent change form, see: Friedman 1987 https://miltonfriedman.hoover.org/internal/media/dispatcher/214346/full

[3] European Central Bank Statistical Data Warehouse. Monetary aggregate M3 vis-a-vis euro area non-MFI excl. central gov. https://sdw.ecb.europa.eu/quickview.do?SERIES_KEY=117.BSI.M.U2.Y.V.M30.X.1.U2.2300.Z01.E

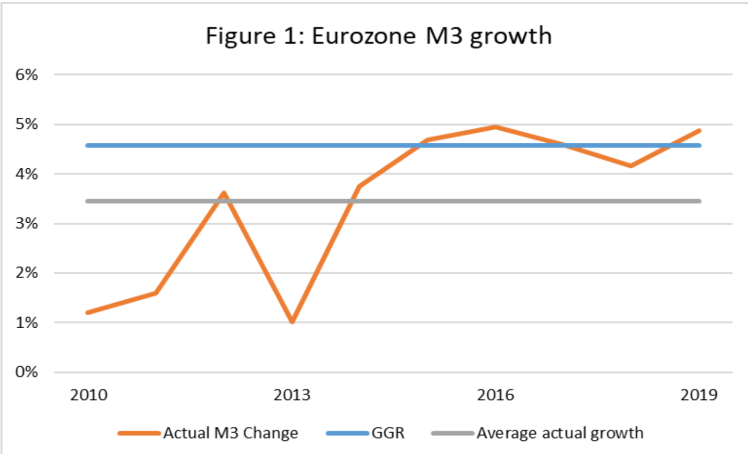
[4] European Central Bank Statistical Data Warehouse. HICP - Overall index. https://sdw.ecb.europa.eu/quickview.do?SERIES_KEY=122.ICP.M.U2.N.000000.4.ANR

[5] Eurostat. Real GDP growth rate – volume. https://ec.europa.eu/eurostat/databrowser/view/tec00115/default/table?lang=en

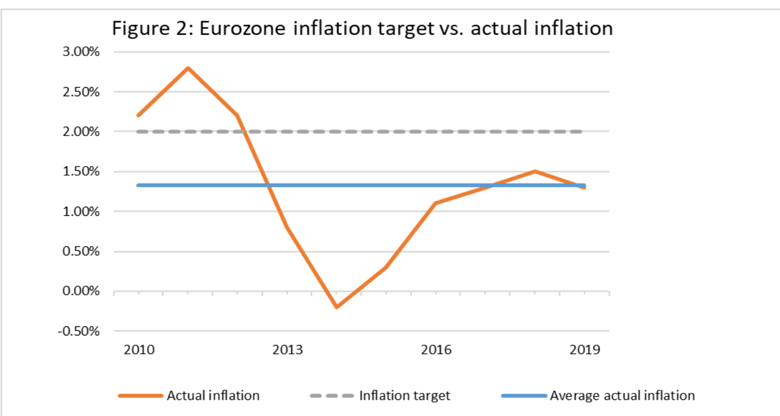
[6] European Central Bank Statistical Data Warehouse. Gross domestic product at market prices. https://sdw.ecb.europa.eu/quickview.do?SERIES_KEY=320.MNA.Q.N.I8.W2.S1.S1.B.B1GQ._Z._Z._Z.EUR.V.N

$1.4\% - (-1.2\%) = 4.6\%$ per year. The actual M3 growth in the eurozone was 3.4% per year, suggesting that **the ECB has been increasing M3 at a lower rate than its golden growth rate**. Not surprisingly, the eurozone's average actual annual inflation between 2010-2019 (1.3%) was lower than the ECB's target of 2%.

The graphs below compare the eurozone's golden growth rate with its actual money supply growth (Figure 1), and its inflation target with its actual inflation level (Figure 2).



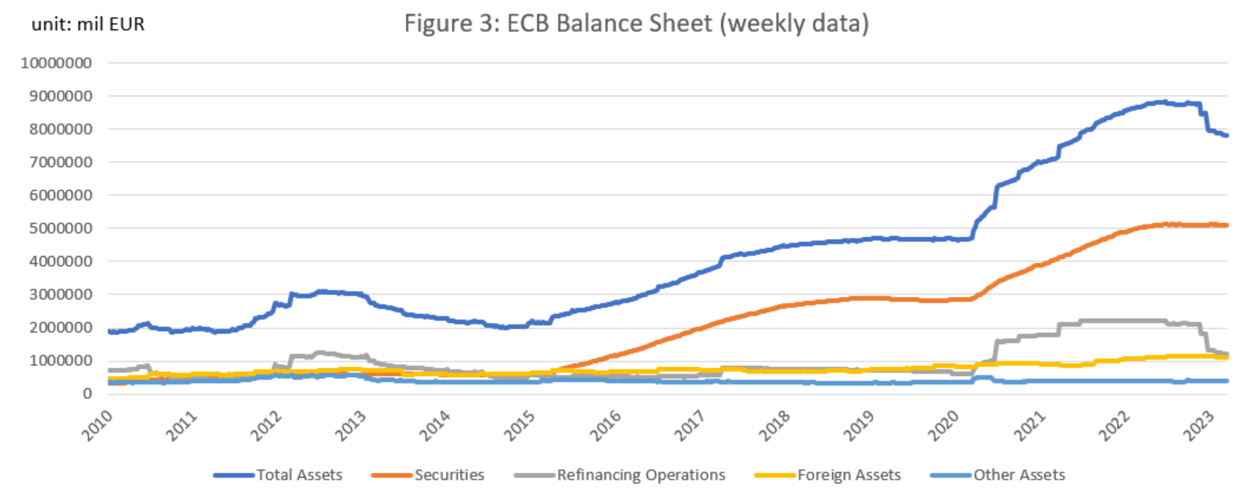
Source: European Central Bank Statistical Data Warehouse, Eurostat
Calculations by Professor Steve H. Hanke



Source: European Central Bank Statistical Data Warehouse, Eurostat
Calculations by Professor Steve H. Hanke

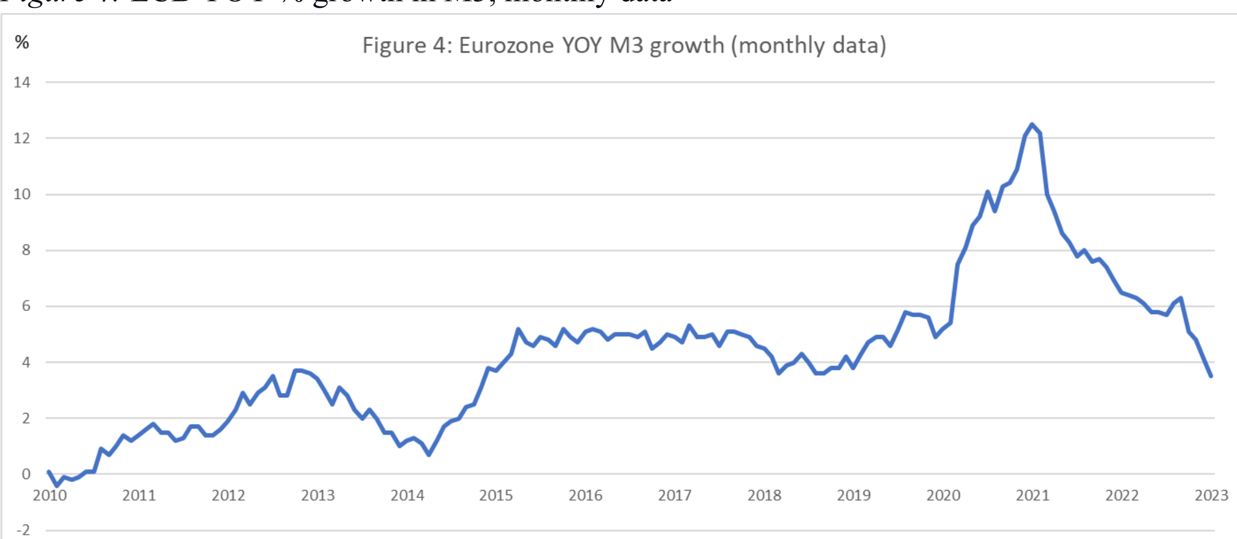
Section 2: ECB's Monetary Operations

To gain a full understanding of the ECB's monetary policies and provide a necessary background for the changes in the money supply, we have analyzed the ECB's balance sheet. Its changes in composition from 2010 through 2022 are presented in Figure 3 below. Figure 4 presents the YoY percentage changes in the eurozone's M3. Together, Figure 3 and Figure 4 help understand the ECB's operations and their effects on the money supply.



Source: European Central Bank
Prepared by Professor Steve H. Hanke

Figure 4: ECB YOY % growth in M3, monthly data



Source: European Central Bank[7]
Prepared by Professor Steve H. Hanke

[7] European Central Bank Statistical Data Warehouse. Monetary aggregate M3 vis-a-vis euro area (index). https://sdw.ecb.europa.eu/quickview.do?SERIES_KEY=117.BSI.M.U2.Y.V.M30.X.I.U2.2300.Z01.A

The ECB's monetary operations between 2010-2022 can be divided into three stages:

1. **Longer-term refinancing operations (LTROs)**, 2011-2015[8]
   During this period, the ECB lent to banks *without* requiring them to make further loans to companies. LTROs consisted of two operations, each with a maturity of 36 months and with the option of early repayment after one year. The increase in ECB lending is reflected through the increase in the ECB's "refinancing operations" in Figure 3. Yet, since the banks were not required to make further (new) loans, its impact on the money supply was minimal; as Figure 4 shows, at best M3 growth reached about 4%/yr for some months in 2012, but quickly dropped close to 0%/yr by 2014, significantly below the eurozone's golden growth rate of 4.58%/yr.
2. **Targeted long-term refinancing operations (TLTRO)**, 2014-present[9]
   (1) **TLTRO I**, June 2014 - March 2016
   TLTRO I was mostly a continuation of the LTROs. It consisted of eight quarterly operations.
   (2) **TLTRO II**, March 2016 - March 2019
   In TLTRO II, the ECB set the maturity of its lendings to be four years. The borrowing rates in the quarterly operations could be as low as the interest rate on the ECB's deposit facility (the eurozone overnight interbank rate). However, the ECB still did not require, but *only* encouraged banks to increase loans to non-financial corporations and households; according to the ECB, the more loans commercial banks issue, the "more attractive the rate on their TLTRO II borrowings becomes."[10]
   (3) **TLTRO III**, March 2019 - present
   TLTRO III was different from the previous operations in that it required banks to make new loans to companies as a condition for receiving ECB loans. In addition, the maturity of these loans was changed to three years (with repayment options after two years). During COVID-19, the ECB announced that the borrowing rates under TLTRO III could be *as low as 50 bps below* the deposit facility rate, and as a bank issues more loans, it receives a lower rate from the ECB. Combined with the QE programme that had started in 2015, the increase in the money supply as a result of TLTRO III was significant: about 10%/yr in 2020.
3. **Quantitative easing (QE)**, 2015-present

[8] European Central Bank. "ECB announces measures to support bank lending and money market activity". https://www.ecb.europa.eu/press/pr/date/2011/html/pr111208_1.en.html

[9] Banco de Portugal. "Targeted longer-term refinancing operations (TLTRO)". https://www.bportugal.pt/en/page/polmon-tltro#:~:text=The%20Governing%20Council%20of%20the,thus%20strengthening%20monetary%20policy%20transmission.

[10] European Central Bank. "Targeted longer-term refinancing operations (TLTROs)". https://www.ecb.europa.eu/mopo/implement/omo/tltro/html/index.en.html

In March 2015, in its initial "quantitative easing" programme, the ECB announced that it would purchase €60 billion per month of euro-area bonds issued by Europe's central governments, agencies, and institutions. However, prior to 2020, these QE programmes followed the "non-liquidity enhancing" model, where the ECB would purchase securities from commercial banks, *not* the non-bank public. The ECB's purchase of securities is reflected in Figure 3. It is arguable that the QE programme may have had little impact in *initiating* the increase in the money supply, since the growth in M3 had already started in mid-2014, well before the ECB's QE began in March 2015 (though it most likely did help sustain its growth rate).

SECTION 3: Credit Counterpart Analysis[11]

Credit counterpart analysis determines what causes the money supply to change. In general, the broad money supply (here, M3) is the sum of currencies held by the nonbank public (currencies are a liability of the central bank, but an asset of the nonbank public) plus all deposits of the nonbank public in the banking system (deposits are liabilities of banks, but an asset of the nonbank public)[12]. The counterparts of these monetary liabilities appear on the asset side of banks' balance sheets and are equivalent to the sum of domestic loans and domestic securities, plus the reserves of banks held at the central bank, plus a residual item that accounts for any discrepancy. Expressed in terms of changes, this is:

Liability side (of central banks):

1. Δ Broad Money = Δ Bank deposits + Δ Cash in circulation with the public, and

Asset side (of commercial banks):

2. Δ Broad Money = Δ Commercial Bank Lending + Δ Securities (held by the banks) + Δ Commercial bank reserves at the central bank +/- Δ Others (net)

A credit counterpart analysis examines changes in the money supply (M3) from the asset side. In this memo, both annual and monthly data are used. The annual data gives an overview of the ECB's monetary operations from 2010 through 2022, while the monthly data provide a more detailed analysis of the ECB's policies before and during the COVID-19 pandemic from 2019 through 2022.

The assets of the eurozone's commercial banks are divided as follows:

(1) **Reserves (commercial bank deposits at the central bank)**

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[11] For logic of the Credit Counterpart Analysis, see: Hanke (2017). https://www.e-elgar.com/shop/usd/money-in-the-great-recession-9781784717827.html.

[12] The ECB defines M3 as the sum of: (1) Currency in circulation + (2) overnight deposits + (3) deposits with an agreed maturity of up to two years + (4) deposits redeemable at notice of up to three months + (5) repurchase agreements + (6) money market fund shares/units + (7) debt securities with a maturity of up to two years Source: European Central Bank. Monetary Aggregates. https://www.ecb.europa.eu/stats/money_credit_banking/monetary_aggregates/html/index.en.html

The ECB does not directly use the term "reserves." Instead, since the commercial bank reserves are the difference between the monetary base and currency in circulation, the monthly data for "reserves" are extracted from *"banks' deposits at the ECB"* from Refinitiv Data Stream, and the annual data are from *"liabilities to euro area credit institutions related to MPOs (monetary policy operations) denominated in euro"* from the ECB's Statistical Data Warehouse (SDW).[13]

(2) **Loans**

These include the commercial banks' lendings to all institutions in the eurozone, excluding the eurozone central banks (including the national central banks in each member state).[14]

(3) **Securities**

These include the commercial banks' holdings of securities issued by all institutions in the eurozone, excluding the eurozone central banks (including the national central banks in each member state)[15].

(4) **Residuals**

These are the difference between the money supply (M3) and the previous three categories.

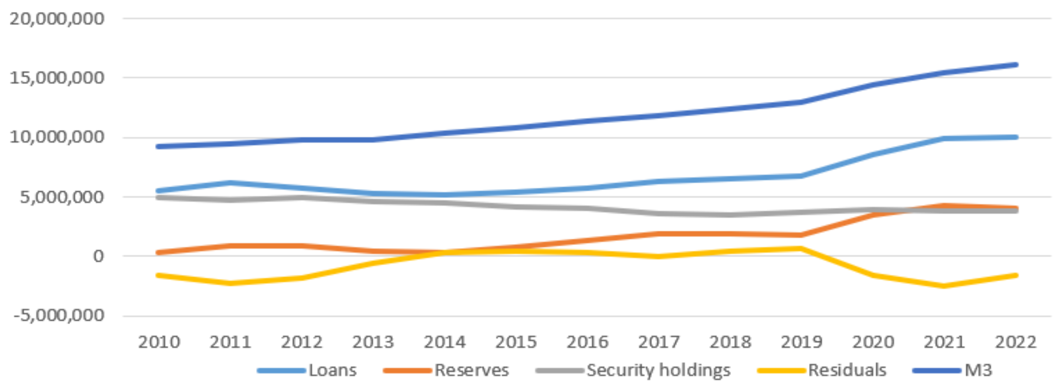
Figure 5 below shows the eurozone's M3 growth and the four items that, in sum, are counterparts that contribute to M3. To better understand how each item contributes to M3 growth, we have also presented Figure 6, which contains M3 and each of its contributors, presented in percentage terms. The sum of these contributions in percentage terms (bank reserves, loans, securities, and residuals) is equal to the annual percentage change in M3. For example, in 2022, M3 grew by 4.4%. The contributors to this change were: bank reserves (0.8%), loans (-1.9%), claims on others (-0.1%), and residuals (5.6%), summing to the M3 growth rate of 4.4%.

[13] European Central Bank Statistical Data Warehouse. Liabilities to euro area credit institutions related to MPOs denominated in euro – Eurosystem. https://sdw.ecb.europa.eu/quickview.do?SERIES_KEY=123.ILM.W.U2.C.L020000.U2.EUR

[14] European Central Bank Statistical Data Warehouse. Loans vis-a-vis euro area MFI. https://sdw.ecb.europa.eu/quickview.do?SERIES_KEY=117.BSI.M.U2.N.A.A20.A.1.U2.1000.Z01.E

[15] European Central Bank Statistical Data Warehouse. Holdings of debt securities reported by MFI excluding ESCB in the euro area. https://sdw.ecb.europa.eu/quickview.do?SERIES_KEY=117.BSI.M.U2.N.A.A30.A.1.U2.0000.Z01.E

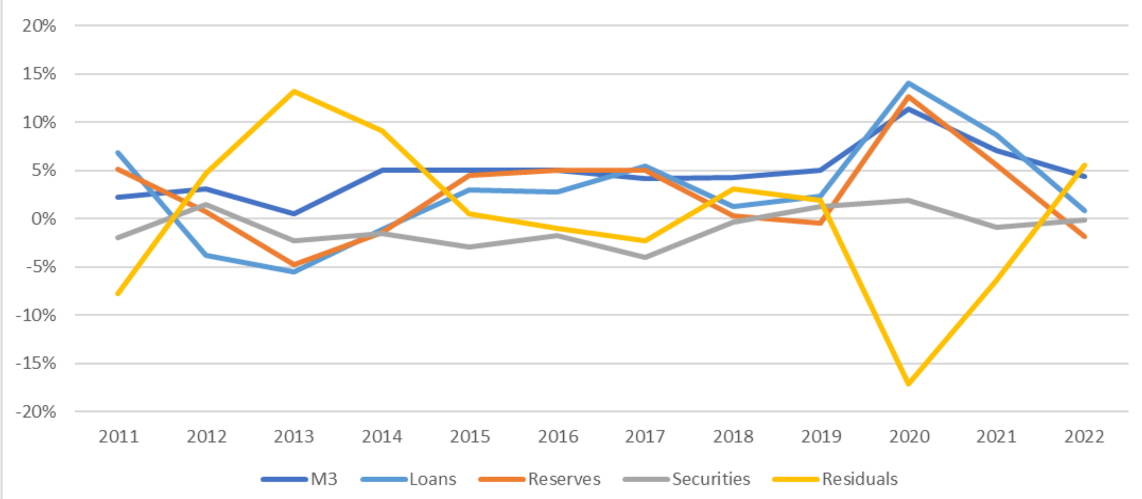
Figure 5: Eurozone Credit Counterpart (annual data)



Unit: million EUR

Source: European Central Bank
Calculations by Professor Steve H. Hanke

Figure 6: Eurozone Credit Counterpart Contributions (annual data)



Source: European Central Bank
Calculations by Professor Steve H. Hanke

Figure 5 shows that the ECB's reserves remained negligible before the ECB announced its first QE programme in 2015. In Figure 6, we also note the negligible/small contribution of commercial bank loans to M3 between 2012-2015 as a result of the ECB's LTRO failing to require additional loans by banks; in fact, loan growth was negative before 2014.

Figures 7 and 8 present the credit counterpart analysis using monthly data for the more recent period of 2019 through 2022. To better understand the ECB's key monetary operations during this period, here is a brief overview[16]:

1. September 18th, 2019: The deposit facility rate was changed from -0.40% to -0.50%.[17]
2. March 18th, 2022: Pandemic Emergency Purchase Programme (PEPP); EUR 750 billion was made available to purchase eligible assets under the existing asset purchase programme (APP):
    (a) bonds purchased directly from banks, which makes more funds available to lend to households or businesses
    (b) purchase of companies' bonds, providing an additional source of credit.
3. June 4th, 2020: PEPP increased from EUR 750 billion to EUR 1.350 trillion
4. June 24th, 2020: The borrowing rate under the TLTRO III operations was decreased; the borrowing rate can now be as low as 50 bps below the deposit facility rate.
5. December 10th, 2020: PEPP increased from EUR 1.350 trillion to EUR 1.850 trillion
6. June 23rd, 2022: The lowest rate for TLTRO III increased to the average interest rate.
7. July 27th, 2022: The deposit facility rate was raised to 0.00% from -0.50%.
8. September 14th, 2022: The deposit facility rate was again raised to 0.75% from 0.00%.

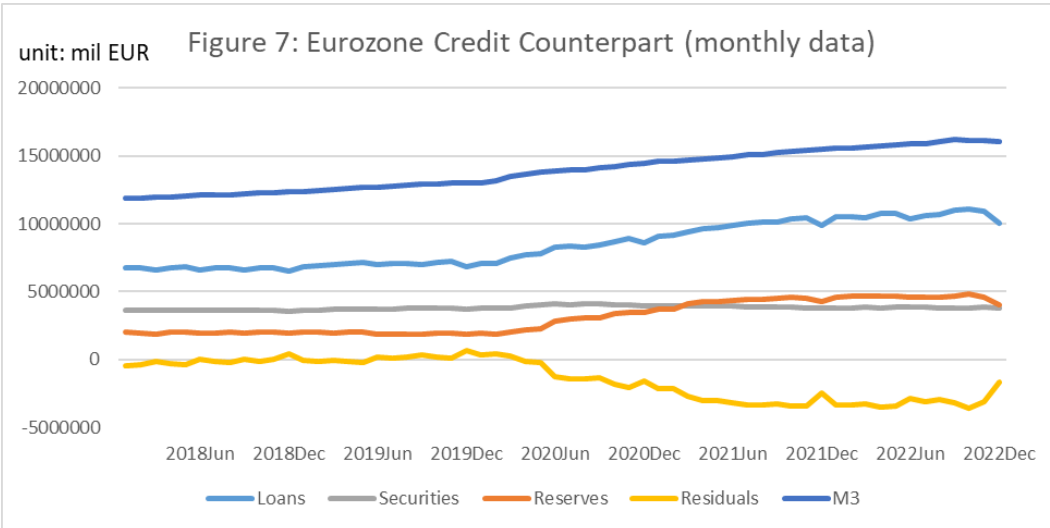
As Figures 7 and 8 show, following the ECB's enactment of the first PEPP (practically a form of quantitative easing) in March 2020, reserves and loans started to increase. The rate of increase accelerated following each increase in the amount purchased under the PEPP (June and December 2020), peaking in December 2020 at a 15% YoY growth rate in reserves. According to the ECB, under these programmes, it purchased bonds from both companies and banks, a move different from its previous QE operations. This, together with its TLTRO operations that required banks to increase lending, stoked M3 to increase at 10% YoY in late 2020 and explains the large increase in the *contribution* of loans to M3 in 2020 in Figure 8.

On 27th July 2022, the ECB, for the first time since 2013, lifted the deposit facility rate (the eurozone overnight interbank rate) out of negative territory to 0%. It further increased it by 75 bps on 14th September 2022. This marked the first time since July 2012 that the rate was positive. As we see in Figure 6, following this, the loan growth rate declined precipitously around September 2022. In absolute terms, the quantity also started to decrease close to December 2022. Consequently, residuals, which are the difference between the money supply and reserves, loans, and securities, started to increase again. However, as Figure 3 shows, the securities held by the ECB have shown no sign of decrease, meaning that it had not yet begun quantitative tightening

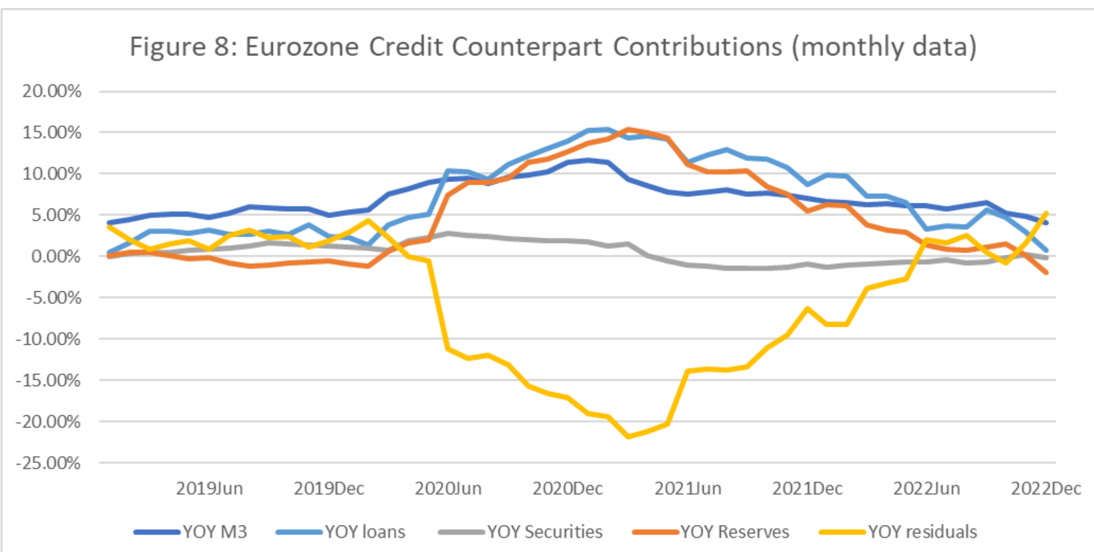
[16] European Central Bank. "Our response to the coronavirus pandemic". https://www.ecb.europa.eu/home/search/coronavirus/html/index.en.html

[17] European Central Bank Statistical Data Warehouse. "Key ECB interest rates". https://www.ecb.europa.eu/stats/policy_and_exchange_rates/key_ecb_interest_rates/html/index.en.html

(QT). Yet, as commercial banks began repaying their loans from the ECB (note that loans under TLTRO had a three-year maturity), the percent increases in M3 started to decline.



Source: European Central Bank
Calculations by Professor Steve H. Hanke



Source: European Central Bank, Refinitiv Data Stream
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